

Supplementary Material for Generalized Emergency Index: A Multi-Trajectory Risk Quantification Framework for Vehicle–Powered Two-Wheeler Interactions

Hao Cheng, Xiaojing Qi, Wenhao Yu, Rui Zhou, Jiang Bian, Yanbo Jiang,
Zhenhua Xu, Heye Huang, Jianqiang Wang, and Sifa Zheng

This document accompanies the main article and supplies extended formulations, proofs, numerical procedures, evaluation details, and additional figures. Sections, equations, figures, tables, propositions, and pages use an S prefix. References to the main article are explicitly identified; the reference list here is local to this supplement.

GUIDE TO THE SUPPLEMENT

- Section **S1**: probabilistic aggregation, default CV/CTRV propagation, and trajectory-conditioned risk definitions.
- Section **S2**: probabilistic decomposition and consistency proofs.
- Section **S3**: interpolation, contact localization, and numerical risk evaluation.
- Section **S4**: warning conventions and context for the main-article case comparison.
- Section **S5**: stochastic input protocol and representative probabilistic interactions.
- Section **S6**: retained-information analysis and detailed numerical results.
- Section **S7**: serial runtime benchmark.
- Section **S8**: full AV2 replay protocol, statistics, and sensor cases.

S1. EXTENDED FUTURE-SET FORMULATIONS

A. Multi-trajectory GEI over weighted future sets

Using the trajectory-conditioned risk definitions reproduced in Section **S1.C**, multiple plausible futures can be aggregated without changing the underlying risk functional. For a finite weighted future set

$$\mathcal{H} = \{h_1, h_2, \dots, h_K\}, \quad (\text{S1})$$

let $w_h(t)$ denote the weight assigned to future hypothesis h , satisfying

$$w_h(t) \geq 0, \quad \sum_{h \in \mathcal{H}} w_h(t) = 1. \quad (\text{S2})$$

The Generalized Emergency Index (GEI) is defined by

$$\text{GEI}(t) = \sum_{\substack{h \in \mathcal{H} \\ w_h(t) > 0}} w_h(t) EI^{(h)}(t). \quad (\text{S3})$$

Equation **(S3)** is the common GEI operator used throughout this work. The trajectory-conditioned quantity $EI^{(h)}$ evaluates the risk associated with one prescribed joint future, whereas the future set and its weights specify how multiple plausible futures are represented. Thus, future generation and weighting are separated from the trajectory-level risk functional.

When plausible futures are represented probabilistically, let

$$H \sim p(H \mid \mathbf{X}_t), \quad (\text{S4})$$

where $\mathbf{X}_t$ denotes the information available at the evaluation instant. For currently separated interactions, Eq. **(S3)** takes the expectation form

$$\text{GEI}(t) = \mathbb{E} [EI^{(H)}(t) \mid \mathbf{X}_t]. \quad (\text{S5})$$

Current geometric contact is handled separately as infinite instantaneous risk, consistently with Eq. **(S33)**.

For a finite discrete future distribution, Eq. **(S5)** becomes

$$\text{GEI}(t) = \sum_{h \in \mathcal{H}} P(H = h \mid \mathbf{X}_t) EI^{(h)}(t). \quad (\text{S6})$$

If $H_1, \dots, H_N$ are independently and identically distributed samples from $p(H \mid \mathbf{X}_t)$,

$$H_n \stackrel{\text{i.i.d.}}{\sim} p(H \mid \mathbf{X}_t), \quad n = 1, \dots, N, \quad (\text{S7})$$

then

$$\widehat{\text{GEI}}(t) = \frac{1}{N} \sum_{n=1}^N EI^{(H_n)}(t) \quad (\text{S8})$$

is the corresponding Monte Carlo estimator. The future probabilities are represented implicitly by the sampling distribution rather than by explicit per-sample weights. GEI itself requires no independence between the future motions of road users A and B . For the Monte Carlo estimator above, only the joint-future samples H_n are assumed to be i.i.d. draws from $p(H \mid \mathbf{X}_t)$. Once a joint future is supplied, its trajectory-conditioned EI is evaluated directly.

B. Default CV/CTRV future configuration

For the default GEI configuration, each road user is assigned two kinematic future hypotheses: constant velocity (CV) and constant turn rate and velocity (CTRV). Their Cartesian product yields four joint future-motion hypotheses.

For each road user $i \in \{A, B\}$, the observed state at time t is

$$\mathbf{S}_i(t) = [x_i(t), y_i(t), v_i(t), \theta_i(t), \omega_i(t), l_i, w_i]^T, \quad (\text{S9})$$

where $x_i(t)$ and $y_i(t)$ denote the Cartesian coordinates of the geometric center, $v_i(t)$ is speed magnitude, $\theta_i(t)$ is heading angle, $\omega_i(t)$ is yaw rate, and l_i and w_i denote body length and width. The current position and velocity vectors are

$$\mathbf{P}_i(t) = \begin{bmatrix} x_i(t) \\ y_i(t) \end{bmatrix}, \quad \mathbf{v}_i(t) = \begin{bmatrix} v_i(t) \cos \theta_i(t) \\ v_i(t) \sin \theta_i(t) \end{bmatrix}. \quad (\text{S10})$$

The resulting joint hypothesis set is

$$\mathcal{H}_{2 \times 2} = \left\{ (\text{CV}, \text{CV}), (\text{CV}, \text{CTRV}), (\text{CTRV}, \text{CV}), (\text{CTRV}, \text{CTRV}) \right\}, \quad (\text{S11})$$

where the first and second entries specify the motion hypotheses of road users A and B , respectively.

Under CV motion,

$$\theta_i^{(\text{CV})}(t + \tau) = \theta_i(t), \quad \mathbf{P}_i^{(\text{CV})}(t + \tau) = \mathbf{P}_i(t) + \tau \mathbf{v}_i(t). \quad (\text{S12})$$

Under CTRV motion,

$$\theta_i^{(\text{CTRV})}(t + \tau) = \theta_i(t) + \omega_i(t)\tau, \quad (\text{S13})$$

and

$$\begin{aligned} \phi_i(t, \tau) &= \theta_i(t) + \omega_i(t)\tau, \\ q_{x,i}(t, \tau) &= x_i(t) + \frac{v_i(t)}{\omega_i(t)} [\sin \phi_i(t, \tau) - \sin \theta_i(t)], \\ q_{y,i}(t, \tau) &= y_i(t) - \frac{v_i(t)}{\omega_i(t)} [\cos \phi_i(t, \tau) - \cos \theta_i(t)], \\ \mathbf{P}_i^{(\text{CTRV})}(t + \tau) &= \begin{cases} [q_{x,i}(t, \tau), q_{y,i}(t, \tau)]^T, & \omega_i(t) \neq 0, \\ \mathbf{P}_i(t) + \tau \mathbf{v}_i(t), & \omega_i(t) = 0. \end{cases} \end{aligned} \quad (\text{S14})$$

Thus, CTRV continuously approaches CV as $\omega_i(t) \rightarrow 0$. For sufficiently small $|\omega_i(t)|$, the numerically stable CV limit or an equivalent local expansion is used.

No externally calibrated preference among the four joint kinematic futures is assumed in the default configuration. Accordingly,

$$w_h = \frac{1}{4}, \quad h \in \mathcal{H}_{2 \times 2}, \quad (\text{S15})$$

and GEI becomes

$$\text{GEI}(t) = \frac{1}{4} \sum_{h \in \mathcal{H}_{2 \times 2}} EI^{(h)}(t). \quad (\text{S16})$$

These uniform weights specify the default future ensemble and need not be interpreted as empirically calibrated maneuver probabilities.

For mathematical representation, the same uniform weighting can equivalently be written as a discrete distribution with equal probability mass:

$$P(H = h | \mathbf{X}_t) = \frac{1}{4}, \quad \forall h \in \mathcal{H}_{2 \times 2}. \quad (\text{S17})$$

This representation does not imply that the four weights are empirically calibrated maneuver probabilities. Substituting

Eq. (S17) into Eq. (S6) recovers Eq. (S16), showing that the default CV/CTRV future set is a special discrete weighted case of the general GEI formulation.

C. Trajectory-conditioned geometry and risk definitions

For completeness, the continuous definitions used in the proofs and numerical procedures are reproduced here. At evaluation time t , a joint future hypothesis h prescribes the center positions $\mathbf{P}_i^{(h)}(t + \tau)$ and headings $\theta_i^{(h)}(t + \tau)$ for $i \in \{A, B\}$ and $\tau \in [0, T]$. These trajectories are continuous and piecewise continuously differentiable, and all admissible hypotheses share the observed bodies at $\tau = 0$. For finite-contact hypotheses, the limiting pre-contact geometry and collision-approach direction are assumed to be well defined.

a) *Occupied bodies and first contact*: For body length l_i and width w_i , the occupied rectangle is

$$\begin{aligned} \mathcal{B}_i^{(h)}(t + \tau) &= \left\{ \mathbf{P}_i^{(h)}(t + \tau) + \alpha \mathbf{e}_i^{(h)}(t + \tau) \right. \\ &\quad \left. + \beta \mathbf{e}_i^{(h)\perp}(t + \tau) \mid \alpha \in \left[-\frac{l_i}{2}, \frac{l_i}{2} \right], \right. \\ &\quad \left. \beta \in \left[-\frac{w_i}{2}, \frac{w_i}{2} \right] \right\}, \end{aligned} \quad (\text{S18})$$

where the heading-aligned basis vectors are

$$\begin{aligned} \mathbf{e}_i^{(h)}(t + \tau) &= \begin{bmatrix} \cos \theta_i^{(h)}(t + \tau) \\ \sin \theta_i^{(h)}(t + \tau) \end{bmatrix}, \\ \mathbf{e}_i^{(h)\perp}(t + \tau) &= \begin{bmatrix} -\sin \theta_i^{(h)}(t + \tau) \\ \cos \theta_i^{(h)}(t + \tau) \end{bmatrix}. \end{aligned} \quad (\text{S19})$$

The 90° rotation operator is

$$\mathbf{J} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad (\text{S20})$$

and the center velocity and yaw rate along the prescribed motion are

$$\mathbf{v}_i^{(h)}(t + \tau) = \frac{d}{d\tau} \mathbf{P}_i^{(h)}(t + \tau), \quad \omega_i^{(h)}(t + \tau) = \frac{d}{d\tau} \theta_i^{(h)}(t + \tau). \quad (\text{S21})$$

The trajectory-conditioned Time for Evasive Maneuver is the first contact time:

$$\begin{aligned} TEM^{(h)}(t) &= \inf \{ \tau \in [0, T] : \mathcal{B}_A^{(h)}(t + \tau) \\ &\quad \cap \mathcal{B}_B^{(h)}(t + \tau) \neq \emptyset \}. \end{aligned} \quad (\text{S22})$$

The infimum of an empty contact set is taken to be $+\infty$; thus $TEM^{(h)}(t) = +\infty$ when the bodies remain separated throughout the horizon. Current geometric contact gives $TEM^{(h)}(t) = 0$. For $0 < TEM^{(h)}(t) < +\infty$, write $\tau_c^{(h)}(t) = TEM^{(h)}(t)$.

b) *Contact-witness geometry*: At a separated instant $\tau < \tau_c^{(h)}(t)$, define the closest-point set by

$$\mathcal{W}^{(h)}(t + \tau) = \arg \min_{\substack{\mathbf{q}_A \in \mathcal{B}_A^{(h)}(t + \tau), \\ \mathbf{q}_B \in \mathcal{B}_B^{(h)}(t + \tau)}} \|\mathbf{q}_B - \mathbf{q}_A\|. \quad (\text{S23})$$

A canonical representative is chosen as

$$(\mathbf{q}_A^{*,(h)}(t + \tau), \mathbf{q}_B^{*,(h)}(t + \tau)) \in \mathcal{W}^{(h)}(t + \tau). \quad (\text{S24})$$

For a nonunique closest-point set, the same deterministic convention selects a representative of the limiting contact feature.

The rigid-body material velocity at the selected point is

$$\begin{aligned} \mathbf{u}_i^{*,(h)}(t+\tau) &= \mathbf{v}_i^{(h)}(t+\tau) \\ &\quad + \omega_i^{(h)}(t+\tau) \mathbf{J} \mathbf{r}_i^{*,(h)}(t+\tau), \\ \mathbf{r}_i^{*,(h)}(t+\tau) &= \mathbf{q}_i^{*,(h)}(t+\tau) - \mathbf{P}_i^{(h)}(t+\tau). \end{aligned} \quad (\text{S25})$$

The limiting relative contact-witness velocity is

$$\mathbf{u}_{BA,c}^{(h)}(t) = \lim_{\tau \rightarrow \tau_c^{(h)}(t)^-} \left[\mathbf{u}_B^{*,(h)}(t+\tau) - \mathbf{u}_A^{*,(h)}(t+\tau) \right]. \quad (\text{S26})$$

When this vector has nonzero norm, the collision-approach direction and tangential projection axis are

$$\mathbf{e}_c^{(h)}(t) = \frac{\mathbf{u}_{BA,c}^{(h)}(t)}{\|\mathbf{u}_{BA,c}^{(h)}(t)\|}, \quad (\text{S27})$$

and

$$\mathbf{n}_c^{(h)}(t) = \mathbf{J} \mathbf{e}_c^{(h)}(t). \quad (\text{S28})$$

The axis $\mathbf{n}_c^{(h)}(t)$ is fixed over the entire pre-contact interval. Numerical treatment of degenerate directions is given in Section S3; a case without a stable direction is marked numerically invalid.

c) *Interaction Depth and Emergency Index:* For $\tau \in [0, \tau_c^{(h)}(t)]$, the tangential center distance is

$$D_c^{T,(h)}(t+\tau) = \left| \left(\mathbf{P}_B^{(h)}(t+\tau) - \mathbf{P}_A^{(h)}(t+\tau) \right)^T \mathbf{n}_c^{(h)}(t) \right|. \quad (\text{S29})$$

The projected body half-width is

$$d_i^{(h)}(t+\tau) = \max_{\mathbf{q} \in \mathcal{B}_i^{(h)}(t+\tau)} \left| \left(\mathbf{q} - \mathbf{P}_i^{(h)}(t+\tau) \right)^T \mathbf{n}_c^{(h)}(t) \right|. \quad (\text{S30})$$

With an optional tangential clearance margin $D_{\text{safe}} \geq 0$, the instantaneous intrusion is

$$D_{\text{in}}^{(h)}(t+\tau) = d_A^{(h)}(t+\tau) + d_B^{(h)}(t+\tau) - D_c^{T,(h)}(t+\tau) + D_{\text{safe}}. \quad (\text{S31})$$

Using $[z]_+ = \max\{z, 0\}$, the trajectory-conditioned Interaction Depth for a finite-contact hypothesis is

$$\text{InDepth}^{(h)}(t) = \max_{\tau \in [0, \tau_c^{(h)}(t)]} \left[D_{\text{in}}^{(h)}(t+\tau) \right]_+. \quad (\text{S32})$$

If no contact occurs within the horizon, set $\text{InDepth}^{(h)}(t) = 0$.

The trajectory-conditioned Emergency Index, including its boundary conventions, is

$$EI^{(h)}(t) = \begin{cases} +\infty, & TEM^{(h)}(t) = 0, \\ \frac{\text{InDepth}^{(h)}(t)}{TEM^{(h)}(t)}, & 0 < TEM^{(h)}(t) < +\infty, \\ 0, & TEM^{(h)}(t) = +\infty. \end{cases} \quad (\text{S33})$$

The primary experiments use $D_{\text{safe}} = 0$, so that no additional clearance margin is imposed beyond the physical body extents. In this case, positive $D_{\text{in}}^{(h)}$ represents tangential projection

overlap without an additional safety buffer. These definitions depend on the supplied body trajectories and the shared witness convention, and do not require a particular future generator.

S2. PROOFS AND CONSISTENCY PROPERTIES

A. Probabilistic decomposition

For currently separated bodies, C_T denotes contact within the prediction horizon under the continuous first-contact definition, and $P_{C,\text{cont}}(t) = P(C_T \mid \mathbf{X}_t)$.

Proposition S1 (Probabilistic decomposition of GEI). *For a currently separated interaction with $P_{C,\text{cont}}(t) > 0$,*

$$\text{GEI}(t) = P_{C,\text{cont}}(t) \mathbb{E} \left[EI^{(H)}(t) \mid C_T, \mathbf{X}_t \right]. \quad (\text{S34})$$

If $P_{C,\text{cont}}(t) = 0$, then $\text{GEI}(t) = 0$.

Proof. By Eq. (S33), $EI^{(H)}(t) = 0$ whenever $H \notin C_T$. For $P(C_T \mid \mathbf{X}_t) > 0$, the law of total expectation gives

$$\begin{aligned} \mathbb{E} \left[EI^{(H)}(t) \mid \mathbf{X}_t \right] &= P(C_T \mid \mathbf{X}_t) \\ &\quad \times \mathbb{E} \left[EI^{(H)}(t) \mid C_T, \mathbf{X}_t \right], \end{aligned} \quad (\text{S35})$$

which yields Eq. (S34). If $P(C_T \mid \mathbf{X}_t) = 0$, the sampled future is non-contacting almost surely and therefore has zero EI almost surely. $\square$

B. Relation to EI and MEI

We finally establish several theoretical properties that clarify the relation between the generalized trajectory-conditioned formulation and the preceding EI/MEI formulations.

Proposition S2 (Motion-generator invariance). *Consider two admissible joint future-motion hypotheses h_1 and h_2 . If they induce identical future center-position and heading trajectories for both road users over $[0, T]$, i.e.,*

$$\begin{aligned} \mathbf{P}_i^{(h_1)}(t+\tau) &= \mathbf{P}_i^{(h_2)}(t+\tau), \\ \theta_i^{(h_1)}(t+\tau) &= \theta_i^{(h_2)}(t+\tau), \end{aligned} \quad \forall i \in \{A, B\}, \tau \in [0, T], \quad (\text{S36})$$

and the same canonical contact-witness convention is applied, then

$$\begin{aligned} TEM^{(h_1)}(t) &= TEM^{(h_2)}(t), \\ \text{InDepth}^{(h_1)}(t) &= \text{InDepth}^{(h_2)}(t), \\ EI^{(h_1)}(t) &= EI^{(h_2)}(t). \end{aligned} \quad (\text{S37})$$

Proof. Identical center-position and heading trajectories induce identical oriented-body evolutions through Eq. (S18). They therefore yield the same first-contact time, limiting closest-point geometry, and rigid-body contact-witness velocities by Eqs. (S22)–(S26) under the shared witness convention. Consequently, Eqs. (S29)–(S32) produce identical intrusion profiles and InDepth values, and Eq. (S33) gives identical EI values. $\square$

Proposition S2 establishes that trajectory-conditioned risk is determined by the prescribed future motion itself rather than by the model class used to generate that motion. Kinematic

models, acceleration-dependent models, planners, and external trajectory predictors can therefore share the same risk functional.

Proposition S3 (Reduction under constant translational motion). *Consider a finite-contact hypothesis h for which both road users undergo constant translational motion over the pre-contact interval:*

$$\theta_i^{(h)}(t+\tau) = \theta_i(t), \quad \mathbf{P}_i^{(h)}(t+\tau) = \mathbf{P}_i(t) + \tau \mathbf{v}_i(t), \quad (\text{S38})$$

for $i \in \{A, B\}$ and $\tau \in [0, TEM^{(h)}(t)]$. Then

$$\mathbf{u}_{BA,c}^{(h)}(t) = \mathbf{v}_B(t) - \mathbf{v}_A(t), \quad (\text{S39})$$

and the contact-witness collision-approach direction coincides, up to orientation, with the relative center-velocity direction. Consequently, under the same body geometry, clearance margin, and TEM convention, the contact-witness InDepth and EI reduce to their constant-translational counterparts.

Proof. Under Eq. (S38), $\omega_i^{(h)} = 0$, so the rotational term in Eq. (S25) vanishes and $\mathbf{u}_i^{*,(h)} = \mathbf{v}_i$. Equation (S39) follows immediately. A reversal of axis orientation does not affect Eqs. (S29) and (S30) because both use absolute projections. The resulting InDepth and EI are therefore identical to the constant-translational formulation. $\square$

Proposition S4 (Exact reduction of GEI over the CV/CTRV future set to MEI at zero yaw rate). *Let $\mathcal{H} = \mathcal{H}_{2 \times 2}$ with any normalized weighting*

$$w_h(t) \geq 0, \quad \sum_{h \in \mathcal{H}_{2 \times 2}} w_h(t) = 1.$$

Assume

$$\omega_A(t) = \omega_B(t) = 0, \quad (\text{S40})$$

and let MEI be evaluated using the same prediction horizon, rectangular-body geometry, clearance margin, nonnegative InDepth convention, and pure-translation TEM rule. Then

$$\text{GEI}(t) = \text{MEI}(t). \quad (\text{S41})$$

Proof. Under Eq. (S40), the CTRV evolution in Eqs. (S13) and (S14) reduces exactly to CV. Hence, all four joint hypotheses in $\mathcal{H}_{2 \times 2}$ generate the same constant-translational future evolution. By Propositions S2 and S3, each hypothesis yields the same trajectory-conditioned EI, equal to MEI. Therefore,

$$\text{GEI}(t) = \sum_{h \in \mathcal{H}_{2 \times 2}} w_h(t) \text{MEI}(t) = \text{MEI}(t), \quad (\text{S42})$$

because the weights sum to one. The same equality holds for the current-contact and no-future-contact boundary cases under the shared conventions. $\square$

Together, Propositions S2–S4 establish two key properties of GEI. First, trajectory-conditioned risk is invariant to the mechanism used to generate a prescribed future. Second, under constant nonrotating motion, the generalized contact-witness geometry reduces to the preceding constant-translational formulation, and GEI over the CV/CTRV future set exactly recovers MEI. These results establish that GEI generalizes the EI-based evasive-demand principle while remaining consistent

with the preceding MEI formulation in the corresponding special case.

S3. NUMERICAL REALIZATION OF THE TRAJECTORY-CONDITIONED RISK FUNCTIONAL

This section summarizes the numerical realization of the trajectory-conditioned TEM, InDepth, and EI defined in Section S1.C. The procedure operates directly on prescribed future trajectories and is independent of the model used to generate them. For clarity, the hypothesis superscript h is omitted throughout this section.

A. Continuous representation of discretely supplied trajectories

Suppose the future motion of road user $i \in \{A, B\}$ is supplied as a discrete pose sequence

$$\Gamma_i = \{\tau_k, \mathbf{P}_{i,k}, \theta_{i,k}\}_{k=0}^K, \quad 0 = \tau_0 < \tau_1 < \dots < \tau_K = T. \quad (\text{S43})$$

The heading sequence is first unwrapped to remove 2π discontinuities. For $\tau \in [\tau_k, \tau_{k+1}]$, let

$$\alpha(\tau) = \frac{\tau - \tau_k}{\tau_{k+1} - \tau_k}. \quad (\text{S44})$$

The continuous pose used for risk evaluation is obtained by piecewise-linear interpolation,

$$\mathbf{P}_i(t + \tau) = (1 - \alpha)\mathbf{P}_{i,k} + \alpha\mathbf{P}_{i,k+1}, \quad (\text{S45})$$

and

$$\theta_i(t + \tau) = (1 - \alpha)\tilde{\theta}_{i,k} + \alpha\tilde{\theta}_{i,k+1}, \quad (\text{S46})$$

where $\tilde{\theta}_{i,k}$ denotes the unwrapped heading. The corresponding piecewise derivatives provide the center velocity and yaw rate required by the contact-witness construction, with left-sided values used at interpolation knots.

This representation allows trajectories produced by kinematic models, learned predictors, planners, or other external modules to be evaluated by the same risk functional without fitting them to a particular motion model.

B. First-contact localization

At a future offset τ , separation and contact between the two oriented rectangular bodies are evaluated using the separating axis theorem. Let

$$\mathcal{A}(\tau) = \{\mathbf{e}_A(\tau), \mathbf{e}_A^\perp(\tau), \mathbf{e}_B(\tau), \mathbf{e}_B^\perp(\tau)\} \quad (\text{S47})$$

denote the candidate separating axes. For an axis $\mathbf{a} \in \mathcal{A}(\tau)$, the projected half-extent of road user i is

$$r_i(\mathbf{a}, \tau) = \frac{l_i}{2} |\mathbf{e}_i(\tau)^T \mathbf{a}| + \frac{w_i}{2} |\mathbf{e}_i^\perp(\tau)^T \mathbf{a}|. \quad (\text{S48})$$

Define the maximum separation margin

$$g(\tau) = \max_{\mathbf{a} \in \mathcal{A}(\tau)} \left\{ \left| (\mathbf{P}_B(t + \tau) - \mathbf{P}_A(t + \tau))^T \mathbf{a} \right| - r_A(\mathbf{a}, \tau) - r_B(\mathbf{a}, \tau) \right\}. \quad (\text{S49})$$

The two occupied regions have nonempty intersection if and only if

$$g(\tau) \leq 0. \quad (\text{S50})$$

The numerical objective is to determine the earliest $\tau_c \in [0, T]$ satisfying Eq. (S50). If the bodies are already in contact at $\tau = 0$, then

$$TEM(t) = 0. \quad (\text{S51})$$

Otherwise, the trajectory intervals induced by Eq. (S43) are examined adaptively.

For an interval $I = [\tau_a, \tau_b]$, let $\tau_m = (\tau_a + \tau_b)/2$ and $\delta = (\tau_b - \tau_a)/2$. A conservative upper bound on the displacement of any material point of road user i over a half interval can be written as

$$B_i(I) = \bar{v}_i(I) \delta + R_i \bar{\omega}_i(I) \delta, \quad (\text{S52})$$

where

$$R_i = \frac{1}{2} \sqrt{l_i^2 + w_i^2} \quad (\text{S53})$$

is the circumradius of the rectangular body, and $\bar{v}_i(I)$ and $\bar{\omega}_i(I)$ bound the center speed and absolute yaw rate over I , respectively.

If a separating axis at τ_m has a positive separation margin larger than the maximum relative material-point displacement allowed over the interval, the entire interval can be certified as contact-free and discarded. Otherwise, the interval is subdivided. This prevents the absence of contact at the supplied trajectory samples from being interpreted as proof of separation over the complete interval.

For intervals that cannot be certified as separated, subdivision is continued until the prescribed temporal tolerance is reached. Guard evaluations and local minimization of $g(\tau)$ are then used to identify a transition from separation to contact. The earliest detected transition provides a bracket

$$[\tau_-, \tau_+], \quad g(\tau_-) > 0, \quad g(\tau_+) \leq 0, \quad (\text{S54})$$

which is refined by bisection. The upper endpoint after refinement is used as the numerical estimate

$$TEM(t) \approx \tau_+. \quad (\text{S55})$$

If no contact occurs over $[0, T]$, then

$$TEM(t) = +\infty. \quad (\text{S56})$$

For nonrotating relative motion, the same definition can be evaluated directly using continuous relative-translation geometry. For general curved trajectories, the adaptive procedure above provides the numerical realization of Eq. (S22).

C. Pre-contact witness and collision-anchored direction

Consider a finite-contact case,

$$0 < TEM(t) < +\infty. \quad (\text{S57})$$

The final separated configuration τ_- in Eq. (S54) approximates the left limit of the first-contact geometry. At this configuration, define a canonical closest-point pair

$$(\mathbf{q}_A^*, \mathbf{q}_B^*) \in \arg \min_{\substack{\mathbf{q}_A \in \mathcal{B}_A(t+\tau_-), \\ \mathbf{q}_B \in \mathcal{B}_B(t+\tau_-)}} \|\mathbf{q}_B - \mathbf{q}_A\|. \quad (\text{S58})$$

When the closest-point set is nonunique, such as in parallel edge-edge configurations, a deterministic representative of the limiting contact feature is used.

The rigid-body velocity of witness point $\mathbf{q}_i^*$ is

$$\mathbf{u}_i^* = \mathbf{v}_i(t + \tau_-) + \omega_i(t + \tau_-) \mathbf{J} [\mathbf{q}_i^* - \mathbf{P}_i(t + \tau_-)]. \quad (\text{S59})$$

The collision-approach direction is then approximated by

$$\mathbf{e}_c = \frac{\mathbf{u}_B^* - \mathbf{u}_A^*}{\|\mathbf{u}_B^* - \mathbf{u}_A^*\|}, \quad (\text{S60})$$

and the tangential projection axis is

$$\mathbf{n}_c = \mathbf{J} \mathbf{e}_c. \quad (\text{S61})$$

If the instantaneous relative witness velocity is numerically degenerate, the limiting direction is approximated using pre-contact finite differences of the witness geometry and, when necessary, the limiting separating direction. If no stable direction can be established, the case is treated as numerically invalid rather than being assigned zero risk.

D. Numerical evaluation of InDepth and EI

Once $\mathbf{n}_c$ is determined, it is fixed over the complete pre-contact interval. For any $\tau \in [0, TEM(t)]$, the instantaneous tangential intrusion is evaluated as

$$D_{\text{in}}(t + \tau) = d_A(t + \tau) + d_B(t + \tau) - \left| (\mathbf{P}_B(t + \tau) - \mathbf{P}_A(t + \tau))^T \mathbf{n}_c \right| + D_{\text{safe}}, \quad (\text{S62})$$

where

$$d_i(t + \tau) = \max_{\mathbf{q} \in \mathcal{B}_i(t + \tau)} \left| (\mathbf{q} - \mathbf{P}_i(t + \tau))^T \mathbf{n}_c \right|. \quad (\text{S63})$$

The numerical Interaction Depth is obtained from

$$\text{InDepth}(t) = \max_{\tau \in [0, TEM(t)]} [D_{\text{in}}(t + \tau)]_+. \quad (\text{S64})$$

The search grid contains the supplied trajectory knots and the refined first-contact time. Additional temporal refinement is applied around the current maximizer of Eq. (S64) to reduce discretization error. The trajectory-conditioned EI is subsequently computed as

$$EI(t) = \frac{\text{InDepth}(t)}{TEM(t)}, \quad 0 < TEM(t) < +\infty. \quad (\text{S65})$$

The boundary conventions follow the complete EI definition in Eq. (S33):

$$\begin{cases} EI(t) = +\infty, & TEM(t) = 0, \\ EI(t) = 0, & TEM(t) = +\infty. \end{cases} \quad (\text{S66})$$

The primary experiments use $D_{\text{safe}} = 0$, so that no additional clearance margin is imposed beyond the physical body extents. In this case, positive D_{in} represents tangential projection overlap without an additional safety buffer. These procedures numerically approximate the trajectory-conditioned definitions for the supplied interpolated motion. Their accuracy depends on the trajectory sampling, contact-search tolerances, and stability of the limiting witness geometry; no general convergence rate or error bound is established here.

S4. EXTENDED RESULTS FOR THE DEFAULT CONFIGURATION

A. Sustained warning distributions

The main article reports signed median warning times. Main-article Figure 3 complements that table with the full distributions. Thresholds are calibrated separately from the 90th, 95th, 99th, and 99.5th percentiles of event-maximum risk over 1,766 non-crash interactions. Warnings must remain active through the final pre-collision segment, with a 0.2 s terminal-boundary tolerance; cases without a valid sustained warning contribute zero lead time. The plotted convention uses the positive warning lead time $t_m - t^*$, for which larger values indicate earlier warnings, whereas the main-article table reports the equivalent signed warning time $t^* - t_m$, for which earlier warnings appear as more negative values.

B. Fine-grained risk differentiation under similar temporal proximity

Main-article Fig. 4 presents the two selected interactions from CIMSS-TA and SIND. Relative to their respective evaluation instants, interaction A develops into a collision after 2.2 s; interaction B is resolved within 0.7 s and reaches a non-interactive state after 2.2 s. The pair illustrates the evasive-demand differences represented by GEI at similar TTC2D, rather than independently measuring the physical avoidance action required.

S5. PROBABILISTIC FUTURE INPUTS AND ADDITIONAL INTERACTIONS

We next evaluate GEI when its plausible future set is supplied by a stochastic multi-trajectory predictor. Motivated by the maneuver-conditioned probabilistic trajectory-sampling structure of Schreier et al. [1], we use a deliberately lightweight stochastic predictor to examine GEI under sampled future distributions. The predictor contains five interpretable motion modes—straight, brake, left turn, right turn, and free motion. Mode priors and Gaussian acceleration/yaw-rate parameters are fitted separately for vehicles and PTWs on a frozen event-level training partition of the conflict corpus (1,350 SIND files and 44 Argo2-LCR files). At the evaluation instant, a Naive-Bayes posterior over these modes is obtained from the currently observed longitudinal acceleration and yaw rate; a mode, acceleration, and yaw rate are then sampled for each actor, followed by exact constant-acceleration/constant-turn-rate propagation over a 3.0 s horizon. In this stochastic predictor, vehicle and PTW futures are sampled independently and paired by Monte Carlo index. The held-out partition is defined deterministically at the source-event level, so neither illustrated interaction contributes data to predictor fitting. This lightweight predictor is used only as a transparent probabilistic input to GEI and is not presented as a high-fidelity reproduction or a benchmark of trajectory-prediction accuracy.

For each illustrative interaction in main-article Fig. 6, the predictor generates $N = 20,000$ joint vehicle-PTW future trajectories. For every sampled joint future $H_n = (\Gamma_A^{(n)}, \Gamma_B^{(n)})$,

the same trajectory pair is used to evaluate collision occurrence and trajectory-conditioned EI; no CV/CTRV futures are introduced.

For comparison with probabilistic temporal criticality, collision probability is evaluated both on a 0.1 s reporting grid and under the refined continuous first-contact semantics used by trajectory-conditioned EI:

$$\hat{P}_{C,0.1s} = \frac{1}{N} \sum_{n=1}^N I_{C,0.1s}(H_n), \quad (S67)$$

$$\hat{P}_{C,\text{cont}} = \frac{1}{N} \sum_{n=1}^N I_{C,\text{cont}}(H_n), \quad (S68)$$

$$\widehat{\text{GEI}} = \frac{1}{N} \sum_{n=1}^N EI^{(H_n)}(t). \quad (S69)$$

The corresponding 20% time-to-critical-collision-probability is denoted by $TTCCP_{20}$. The grid-based and continuous-contact probabilities are reported separately because they use different contact-detection semantics.

To determine whether the distinction extends beyond selected cases, we conduct a protocol-frozen population analysis on the pre-existing strictly held-out SIND risk-range pool. The pool was defined without conditional EI by requiring a proxy collision probability in $[0.1, 0.9]$ and a finite proxy mean contact time, yielding 347 interaction instants from 162 source events. Each instant is recomputed using $N = 5,000$ joint futures; conditional analysis further requires finite $TTCCP_{20}$ and at least 100 continuous-contact samples, retaining 222 instants from 118 source events. To avoid pseudo-replication, source events are paired without replacement by maximum-cardinality, minimum-distance matching under $|\Delta \hat{P}_{C,\text{cont}}| \leq 0.02$ and $|\Delta TTCCP_{20}| \leq 0.1$ s, with conditional EI excluded from the matching variables. Uncertainty is assessed using 10,000 pair-level bootstrap replicates; an independent $N = 1,000$ rerun of each matched event quantifies Monte Carlo variation.

Event-level matching yields 56 independent pairs covering 112 distinct source events. Although the matching variables are tightly controlled, the median absolute conditional-EI difference is 0.233 m/s (95% pair-bootstrap CI: 0.174–0.278 m/s), and its 90th percentile is 0.441 m/s. The median difference is 14.9 times the independent-run Monte Carlo median of 0.016 m/s, indicating that the observed separation is not explained by sampling noise alone. The corresponding median absolute $\widehat{\text{GEI}}$ difference is 0.079 m/s, and 12.5% of matched pairs differ by more than a factor of two. Across six pre-specified matching-caliper settings, the median conditional-EI difference remains 0.192–0.236 m/s, with 90th percentiles of 0.441–0.502 m/s. These population results show that, within the frozen predictor, the GEI formulation retains a distinct conditional evasive-demand component after collision likelihood and temporal criticality are controlled; they are not an outcome-prediction comparison.

Main-article Figure 6 provides a geometric view of two interactions marked in main-article Fig. 5 that have closely matched probabilistic temporal criticality. Their grid-based

collision probabilities differ by only 0.00245, their continuous-contact probabilities by 0.01455, and their $TTCCP_{20}$ values by 0.1 s. Nevertheless, $\widehat{GEI}$ increases from 0.22221 to 0.84355 m/s, a factor of approximately 3.80. The corresponding 95% Monte-Carlo confidence intervals do not overlap.

The decomposition in Proposition S1 explains this separation. Among continuously contacting futures, the mean trajectory-conditioned EI increases from 0.30431 to 1.13266 m/s, while the corresponding mean InDepth increases from 0.45929 to 1.81380 m. The mean TEM values are 1.50899 and 1.65390 s, respectively. Thus, although Interaction B has slightly more time before contact on average, its collision-producing futures require substantially greater evasive demand. The difference in GEI is therefore driven primarily by the conditional criticality of contact-producing futures rather than by collision occurrence probability alone.

The A/B pair was selected under the frozen held-out protocol and satisfies the pre-specified collision-probability and $TTCCP_{20}$ matching criteria.

S6. EXTENDED CRASH-OUTCOME INFORMATION ANALYSIS

We next quantify how much information about the eventual crash-versus-non-crash outcome is retained in each method’s instantaneous scalar risk value. This analysis uses the same aligned interaction states for all methods and evaluates GEI against the baseline measures.

The crash set contains 303 reconstructed vehicle–PTW crashes with complete trajectories from -3.0 to -0.1 s before collision. The non-crash set contains 1,605 potential-conflict episodes with complete aligned trajectories over the same interval. Crash cases are aligned by the true crash time ($t = 0$), whereas non-crash episodes are aligned by their minimum bounding-box distance, which provides a method-independent geometric reference. This alignment is used only to define the retrospective reference time and does not enter the computation of any instantaneous risk metric. All episodes are evaluated on $\tau \in \{-3.0, -2.9, \dots, -0.1\}$ s.

For method j , let $M_j(\tau)$ denote its scalar score at reference-relative time τ . At each time slice, the oriented scalar score is mapped to an out-of-fold calibrated crash probability using isotonic regression fitted only on the corresponding training folds. Retained crash-outcome information is defined as the reduction in held-out calibrated cross-entropy relative to the empirical crash-outcome entropy:

$$\widehat{\mathcal{I}}_j^{\text{ret}}(\tau) = H_\tau(Y) - \widehat{CE}_\tau(Y | M_j), \quad (\text{S70})$$

where $Y \in \{0, 1\}$ denotes the eventual crash outcome. Both entropy and cross-entropy are computed using base-2 logarithms and are therefore reported in bits. Because the same 303 crash and 1,605 non-crash episodes are present at every sampled time, the crash base rate is 0.159 and the corresponding baseline entropy is 0.631 bits throughout the analysis. Equation (S70) should be interpreted as held-out log-loss reduction relative to the empirical base-rate predictor rather than as a direct estimator of mutual information.

Main-article Figure 7(a) shows that GEI provides the highest mean AUROC over the full reference-relative horizon. Its

mean AUROC is 0.886, compared with 0.861 for TAdv, 0.802 for 2D-TTC, 0.801 for TTC2D, 0.800 for MEI, and 0.799 for ACT. At the earliest sampled time (-3.0 s), TAdv has the highest AUROC, while GEI is second; the latter becomes the strongest method through most of the subsequent horizon.

The retained-information analysis in main-article Fig. 7(b) provides a complementary view. GEI retains the largest average crash-outcome information, at 0.331 bits or 52.3% of the baseline entropy. The next strongest methods are TTC2D (0.245 bits), ACT (0.243 bits), MEI (0.238 bits), and 2D-TTC (0.233 bits).

Table S1 shows that the advantage of GEI is concentrated in the early and intermediate portions of the horizon. At -2.5 s, it retains 0.114 bits, compared with 0.089 bits for TAdv and 0.040 bits for TTC2D. At -1.5 s, it retains 0.362 bits, substantially above TAdv (0.226 bits), TTC2D (0.207 bits), and MEI (0.201 bits). Near the reference event, ACT and TTC2D slightly exceed GEI at -0.1 s, consistent with the earlier observation that proximity-based measures become highly informative close to collision.

Overall, GEI retains more crash-outcome-relevant information in its scalar score across most of the pre-event horizon, particularly before the final near-collision phase.

S7. SERIAL COMPUTATIONAL EFFICIENCY

We next evaluate the computational cost of the default GEI configuration. Frame-level computation is performed over the 1,766 conflict cases under serial execution on a Windows 10 workstation equipped with an Intel Core i9-13900HX processor and 32.0 GB RAM. The Python implementation uses NumPy 1.26.4 and pandas 2.1.4. Computation time is measured using `time.perf_counter()` and excludes file I/O and data-loading overhead. No GPU acceleration is used.

Across 175,053 valid frames, GEI requires a mean computation time of 4.27 ms/frame and a median of 3.99 ms/frame. The P90, P95, and P99 times are 6.82, 9.95, and 18.50 ms/frame, respectively, with a maximum observed time of 40.81 ms/frame. These results indicate that the default GEI configuration has low per-frame computational cost when future trajectories are generated by the simple CV/CTRV hypotheses.

The reported runtime should not be interpreted as the computational cost of arbitrary GEI configurations. For predictor-provided futures, the GEI workload additionally depends on the number, temporal resolution, and geometric complexity of the supplied trajectories.

S8. COMPLETE FRAME-CAUSAL AV2 SENSOR REPLAY

To examine whether GEI can operate sequentially on real on-road tracks and accept heterogeneous future generators without changing its risk functional, we conduct a deployment-oriented frame-causal replay using the Argoverse 2 (AV2) Sensor Dataset [2]. Frame-causal means that the risk score at timestamp t uses only information available at or before t . The study is an open-loop on-road log replay rather than a closed-loop, hardware-in-the-loop, or vehicle-in-the-loop test: the computed risk does not alter the recorded trajectories.

TABLE S1

RETAINED CRASH-OUTCOME INFORMATION AT REPRESENTATIVE REFERENCE-RELATIVE TIMES. PANELS SHOW THE SAME RESULTS IN BITS AND AS PERCENTAGES OF THE BASELINE ENTROPY (0.631 BITS). BOLDFACE MARKS EACH ROW'S LARGEST DISPLAYED VALUE; TIES ARE RETAINED.

(a) Retained information (bits)								
Time	GEI	MEI	ACT	TTC2D	2D-TTC	TTC	TAdv	DRAC
-3.0 s	0.047	0.016	0.003	0.013	0.002	0.002	0.053	0.003
-2.5 s	0.114	0.037	0.031	0.040	0.039	0.007	0.089	0.026
-2.0 s	0.217	0.093	0.101	0.104	0.100	0.031	0.146	0.047
-1.5 s	0.362	0.201	0.194	0.207	0.194	0.077	0.226	0.082
-1.0 s	0.461	0.349	0.352	0.354	0.347	0.145	0.294	0.140
-0.5 s	0.560	0.524	0.542	0.543	0.510	0.177	0.361	0.174
-0.1 s	0.580	0.569	0.587	0.588	0.530	0.095	0.372	0.090
Average	0.331	0.238	0.243	0.245	0.233	0.082	0.214	0.088
(b) Fraction of baseline entropy (%)								
Time	GEI	MEI	ACT	TTC2D	2D-TTC	TTC	TAdv	DRAC
-3.0 s	7.5	2.6	0.5	2.1	0.3	0.2	8.4	0.4
-2.5 s	18.1	5.8	4.9	6.3	6.1	1.1	14.1	4.0
-2.0 s	34.3	14.7	15.9	16.4	15.8	4.9	23.1	7.4
-1.5 s	57.4	31.8	30.7	32.8	30.6	12.2	35.9	13.0
-1.0 s	73.0	55.2	55.8	56.1	55.0	22.9	46.6	22.2
-0.5 s	88.7	83.0	85.8	86.0	80.7	28.0	57.2	27.6
-0.1 s	91.8	90.2	93.0	93.0	83.9	15.0	58.9	14.3
Average	52.3	37.6	38.5	38.8	36.9	12.9	33.8	13.9

The replay covers all 700 training and 150 validation logs for which 3-D annotations are publicly available. Among these 850 logs, 278 contain at least one object labeled MOTORCYCLE; MOTORCYCLIST is excluded to avoid counting the rider and motorcycle as separate physical PTWs. A deterministic registry splits tracks at gaps greater than 0.25 s, yielding 584 continuous ego-PTW events, 53,674 ego-PTW pair-frames, and 31,260 distinct PTW-containing timestamps. No motion, distance, conflict, or risk pre-screening is applied to this registry. All three configurations therefore evaluate exactly the same event-timestamp keys, including stationary PTWs and newly appearing tracks. Independent rereading of the outputs confirms zero missing, duplicate, unexpected, or nonfinite results.

The motorcycle cuboids and 6-DoF ego poses are transformed to a common city frame. For the default GEI configuration, speed and yaw rate are estimated causally: the first observation uses zero-motion initialization, the second uses a backward difference, and later observations use least-squares fitting over at most five current/past samples. For the learned predictors, up to 50 current/past history slots are exposed, but a complete 50-frame history is not required; observed prefixes are right-aligned and unavailable history is masked. Consequently, short histories are evaluated rather than rejected or filled with future annotations. The synchronized camera and LiDAR data are used only for contextual audit and visualization, not as GEI inputs.

Three configurations are replayed separately. The first uses the default CV/CTRV future set and requires no external trajectory predictor. The other two retain all six probability-weighted future modes from EMP-D [3] or QCNet [4] for both

ego and PTW, producing 36 trajectory pairs per interaction over a 3 s, 10-Hz horizon. EMP-D provides a lightweight predictor, whereas QCNet uses query-centric scene encoding and interaction-aware multi-agent context. Both pretrained AV2 Motion Forecasting models are applied directly to AV2 Sensor Dataset tracks without fitting a GEI risk threshold on the replay data. Because AV2 does not provide counterfactual ego-planner outputs, the learned ego modes are interpreted as forecast hypotheses for risk integration, not as executable planning commands. The same GEI operator is used without modification for both future generators.

Main-article Figure 8(a) shows that the event-level distributions are strongly concentrated near zero but retain configuration-dependent upper tails. Across the 584 events, the mean, P99, and maximum peak risks are 0.0415, 0.6357, and 0.9276 m/s for GEI; 0.0353, 0.6666, and 1.0216 m/s for GEI (EMP-D); and 0.0104, 0.2514, and 1.7289 m/s for GEI (QCNet), respectively. These differences reflect the distinct future trajectories and mode probabilities supplied by the three configurations rather than a change in the underlying GEI risk functional.

For transparent qualitative selection, main-article Fig. 9 and Fig. S2 show two moving-PTW events whose interaction-mean PTW speeds exceed 0.5 m/s. Among the 165 events satisfying this motion criterion, AV2S-PTW-0275 ranks 2nd, 5th, and 2nd by event-level peak risk under GEI, GEI (EMP-D), and GEI (QCNet), respectively; AV2S-PTW-0334 ranks 10th, 1st, and 3rd. The use of different future generators naturally changes the numerical risk because the predicted trajectories and mode probabilities differ. Nevertheless, in both examples the three GEI configurations localize elevated

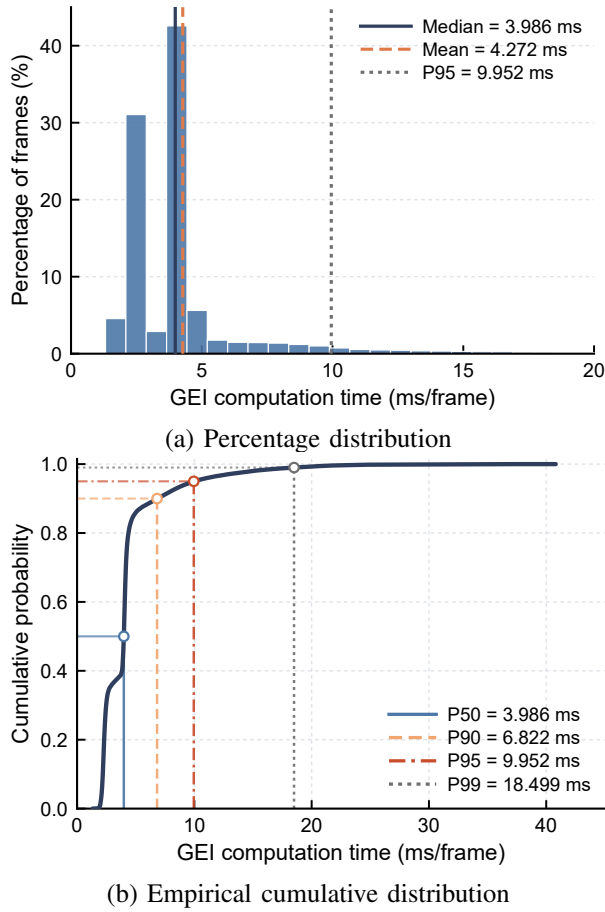


Fig. S1. Per-frame computation time of GEI over 175,053 valid frames from 1,766 conflict cases. (a) Percentage distribution of per-frame computation time; values above 20 ms/frame are omitted for visual clarity, while percentages retain all valid frames as the denominator. (b) Empirical cumulative distribution function (ECDF), with markers for the median, P90, P95, and P99 computation times.

risk to the same short interaction interval, place their peaks within 0.20–0.30 s, and produce peak-risk magnitudes within the same order of magnitude. This case-level commonality reflects the shared GEI risk functional, whereas the residual differences arise from the future distributions supplied by the different generators. It does not imply that the three full-dataset score scales are directly comparable.

Accordingly, the default GEI configuration provides a reproducible baseline when a calibrated multimodal predictor is unavailable. When a reliable predictor is available, GEI can instead exploit scene-dependent, map-aware, and interaction-aware futures to represent a broader range of maneuver hypotheses. The formulation is compatible with different future generators through a common trajectory interface, but its numerical scores are not calibration-free: each predictor–risk configuration requires its own operating calibration before warning deployment.

Runtime is measured on the 4090D server with an NVIDIA GeForce RTX 4090 D; the three configurations are executed as isolated workloads, and the predictor-based pipelines use an optimized eight-core input path. As shown in main-article Fig. 8(b), GEI has a mean/P99/maximum total workload of

5.48/25.96/47.81 ms, remaining below 100 ms for all 31,260 updates. For EMP-D, network inference and the GEI core have P99 workloads of 14.09 and 27.54 ms, and their combined P99 is 39.57 ms; 99.77% of updates remain below 100 ms. For QCNet, the corresponding P99 values are 95.12, 19.33, and 105.49 ms, with 98.73% below 100 ms. Thus, the GEI risk cores themselves have P99 workloads below 28 ms, while the selected future generator determines the remaining real-time margin. The observed tails therefore do not establish strict per-update compliance with the 100 ms interval. Input construction is measured separately and is not included in “combined” time; its P99 is 251.03 ms for EMP-D and 34.29 ms for QCNet.

These timings exclude sensing, detection, tracking, disk I/O, and visualization and therefore are module-level workloads rather than end-to-end vehicle latency. Moreover, AV2 provides neither frame-level hazard labels nor a warning–response feedback loop. The replay therefore demonstrates frame-causal track compatibility, future-generator compatibility, and module-level computational feasibility. It does not establish false-positive or true-positive performance, warning lead time, or closed-loop safety benefit.

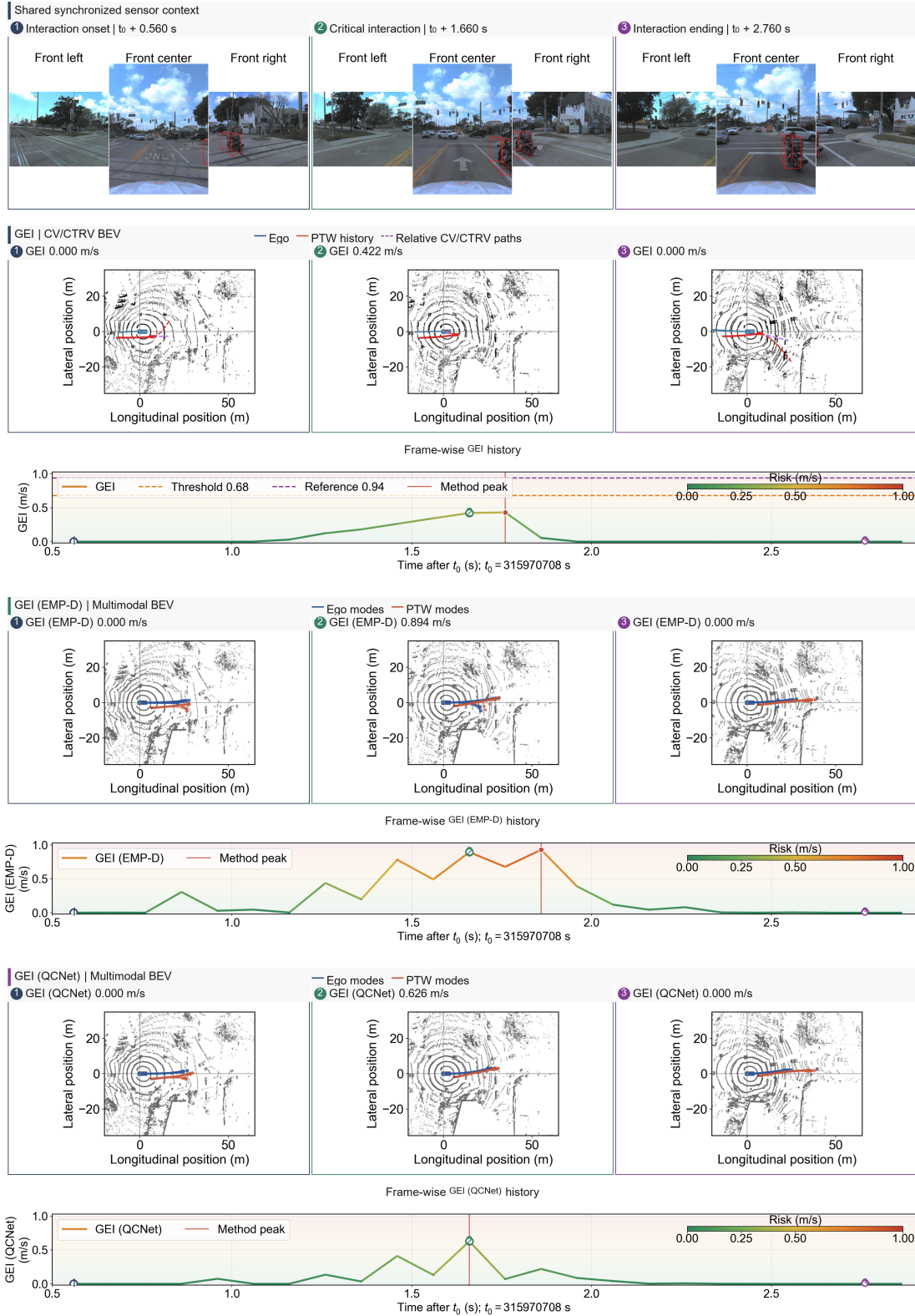


Fig. S2. Frame-causal comparison for AV2S-PTW-0334. Three shared instants combine the synchronized front-left, front-center, and front-right cameras with configuration-specific LiDAR bird's-eye views, future trajectories, and risk histories. Event-level peaks are 0.431 m/s for GEI, 0.928 m/s for GEI (EMP-D), and 0.626 m/s for GEI (QCNet); all occur within a 0.20 s interval.

REFERENCES

- [1] M. Schreier, V. Willert, and J. Adamy, "Bayesian, maneuver-based, long-term trajectory prediction and criticality assessment for driver assistance systems," in *2014 IEEE 17th International Conference on Intelligent Transportation Systems (ITSC)*. IEEE, 2014, pp. 334–341.
- [2] B. Wilson, W. Qi, T. Agarwal, J. Lambert, J. Singh, S. Khandelwal,

- B. Pan, R. Kumar, A. Hartnett, J. K. Pontes *et al.*, “Argoverse 2: Next generation datasets for self-driving perception and forecasting,” *arXiv preprint arXiv:2301.00493*, 2023.
- [3] A. Prutsch, H. Bischof, and H. Possegger, “Efficient motion prediction: A lightweight and accurate trajectory prediction model with fast training and inference speed,” in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2024.
- [4] Z. Zhou, J. Wang, Y.-H. Li, and Y.-K. Huang, “Query-centric trajectory prediction,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 17 863–17 873.